

Table 2: The technical differences between MongoDB and RethinkDB

Differences	MongoDB	RethinkDB
Platforms supported	Linux, Windows, OS X, Solaris	Ubuntu, OS X, Centos OS, Debian Wheezy and Jessie
Data model	Supports the BSON data model. The BSON protocol is an extension to JSON and supports additional data types like ObjectId, time stamp, date time, etc.	Stores JSON documents with a binary on disk serialisation.
Query language interpreter	Data queries can be done via CRUD operations using BSON objects.	Makes use of the unified chainable query language which supports CRUD operations, aggregations, joins and sub-queries.
Support for JavaScript	Full support for JavaScript. MongoDB uses map-reduce functions, which are based on JavaScript. It makes use of the V8 engine.	Supports integration of JavaScript expressions anywhere in the language.
Access languages	Supports 10 official binary languages.	Official languages supported are JavaScript, Node.js, Python, Java and Ruby.
Cloud support	Supports various cloud platforms like AWS, Joyent, Rackspace, etc.	Can be deployed only via manual settings on AWS.
CLI/UI tools	CLI: Simple JavaScript interactive shell for data inspection, testing queries and creating indexes. UI: Simple HTTP interface for read-only information about a server.	CLI: It includes fine-grained cluster and server configuration. UI: Web based admin console is accessible via every node to provide high level and well-guided support.
Backup utility	Backup is done via the <i>mongodump</i> utility.	Supports hot backup on live cluster via <i>dump</i> and <i>restore</i> commands.
Sharding facility	Automatic range-based sharding using shard key; requires three config servers and one or more instances.	Just one-click sharding using the Web based admin console.
Replication	Master-slave replication using asynchronous approach.	It allows replication using one-click admin Web UI or from CLI.
Concurrency	Uses locks at various levels for ensuring data consistency.	Implements block-level multi-version concurrency control.
Storage engine	Makes use of memory mapped files.	Data is organised into B-trees and stored on disks using a log-structured storage engine based on BTRFS architecture.

RethinkDB versus MongoDB

RethinkDB is considered as a good choice for those developers who build real-time applications. When a user makes a request, the application needs to respond at near-instantaneous speeds. Like MongoDB, RethinkDB is a fast and flexible JSON-based database management system.

RethinkDB is no doubt a good database engine but it has certain limitations in terms of administration and development.

So when it comes to the overall rating of RethinkDB as compared to

MongoDB, the latter is better on a few counts (see Table 2). 

References

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